

Co-Producing Environmental Publics: Technology, Communication, and Ecological Transformation

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In recent decades, environmental challenges—ranging from climate change and air pollution to biodiversity loss and resource scarcity—have increasingly shaped not only policy agendas but also the very texture of public life around the globe. Responding to these crises, and simultaneously transforming them, digital technologies—including sensor networks, big data analytics, and artificial intelligence—have become constitutive elements in how environmental issues are rendered knowable and actionable (Olawade et al., 2024; Kloppenburg et al., 2022). These sociotechnical developments do more than document ecological change; they intervene in the communicative infrastructures through which publics emerge, take shape, and act. Technologies of sensing, modeling, and prediction increasingly define what becomes visible or knowable as “environmental risk,” thereby shaping how responsibility, urgency, and agency are understood in confronting ecological crises (Muncie, 2025; Turnbull et al., 2022). Yet these knowledge infrastructures operate unevenly: through algorithmic filtering, platform architectures, and unequal access to data regimes, digital systems intensify longstanding inequalities in visibility, participation, and recognition—particularly in contexts of rapid or uneven environmental degradation—thereby reconfiguring whose experiences, claims, and knowledges enter public discourse (Muncie, 2025; Bolton, Raven & Mintrom, 2021). In this context, digital infrastructures and ecological conditions mutually constitute the communicative possibilities through which publics take form, calling for a re-examination of foundational questions of publicness and communication in a world where ecological and technological conditions increasingly co-produce the possibilities for collective life (Staab & Sorg, 2025; Wickberg et al., 2024; Kloppenburg et al., 2022). Yet existing theories, largely premised on

stable media environments and human-centered deliberation, struggle to account for publics constituted through algorithms, sensors, and predictive ecologies.

Digital technologies have fundamentally altered the mediating structures through which publics encounter and engage with environmental issues, introducing new possibilities for participation while simultaneously imposing novel forms of constraint. Scholars note that digital devices and platforms have enabled more granular, participatory, and interactive forms of environmental knowledge—expanding opportunities for publics to witness environmental change and mobilize around shared concerns (Eilstrup-Sangiovanni & Hall, 2024). Yet these same infrastructures operate as sites of power: algorithmic prioritization determines which environmental claims gain visibility, data regimes structure whose knowledge counts as authoritative, and platform governance decides the terms of public participation (Padden, 2023). As infrastructures of environmental governance, digital systems do not merely represent ecological conditions but actively constitute what counts as valid environmental evidence, which actors are recognized as legitimate knowledge producers, and whose experiences are rendered visible or invisible in decision-making (Kloppenborg et al., 2022). This raises fundamental questions about epistemic authority and communicative power in an era where environmental publics are increasingly mediated by algorithmic systems. Meanwhile, environmental misinformation, AI-generated narratives, and computational propaganda destabilize the epistemic foundations on which publics deliberate ecological futures (Tomassi, Falegnami & Romano, 2025; Wetts et al., 2025). The emergence of hybrid media systems (Chadwick, 2017)—where institutional actors, grassroots communities, and algorithmic intermediaries interact dynamically—further complicates how environmental issues circulate, gain salience, and become contested (Aronczyk et al., 2025). These developments reveal the limited capacity of legacy models to account for how environmental publics are formed, fragmented, or transformed under shifting communicative and ecological conditions.

Against this backdrop, this special issue seeks to advance scholarly understanding of how technological systems reshape environmental communication and how ecological crises, in turn, reconfigure the communicative, institutional, and imaginative infrastructures of public life. By foregrounding the mutually constitutive relationship between technology, publics, and ecological crises, it aims to deepen theoretical debates on public formation, algorithmic governance,

mediated knowledge production, and the rearticulation of authority and collective action in an era of planetary uncertainty. The issue is organized across three interconnected levels. At the macro level, we explore how algorithmic infrastructures and platform governance shape the conditions under which environmental publics form—defining who can participate, which issues gain visibility, and what forms of collective action are possible. At the meso level, we investigate how environmental evidence is generated, circulated, and contested, examining what counts as authoritative knowledge and how epistemic power is distributed across hybrid media ecologies involving scientists, activists, Indigenous communities, and nonhuman actors. At the micro level, we examine communicative agency and collective action—how diverse actors navigate, resist, or transform the technological and ecological conditions shaping their participation in environmental public discourse. This special issue critically explores how emerging digital infrastructures—from sensing technologies and predictive models to platformized media environments—are transforming the circulation of environmental claims and the conditions under which publics coalesce around ecological concerns. Contributors are encouraged to engage with these developments from interdisciplinary perspectives, drawing on communication studies, science and technology studies, environmental governance, political ecology, and related fields. By centering the intersections of technological mediation and ecological transformation, this inquiry aims to illuminate the evolving dynamics that define environmental publics in the digital age, shedding light on how environmental communication is being reconfigured as a site of public struggle, imagination, and governance.

Themes and Scope

We welcome conceptual, methodological, and empirical studies that examine how digital technologies mediate environmental governance, identity formation, activism, and the circulation of ecological knowledge. We particularly encourage research on how digital platforms structure public debate on climate change, how algorithmic systems shape visibility and legitimacy in environmental decision-making, and how environmental risk communication unfolds during crises. Studies situated in underrepresented regions—including Asia, Africa, Latin America, and Indigenous communities—are especially welcome for the ways they challenge Western-centric

assumptions and illuminate alternative modes of public formation under technological and ecological pressure. By assembling these diverse perspectives, the special issue seeks to generate new conceptual and empirical insights into the evolving relationship between communication, technology, and the environment—an urgent scholarly task in light of accelerating ecological transformation and the expanding role of digital systems in structuring public life. Potential topics include, but are not limited to:

- Algorithmic Infrastructures and the Formation of Environmental Publics
- Datafication, Environmental Knowledge, and Public Authority
- Public Communication of Climate Models, Predictive Ecologies, and Digital Simulations
- Networked Environmental Activism and Hybrid Public Mobilization
- Communicative Agency among Scientists, Indigenous Communities, and Climate Advocates
- Surveillance Ecologies, Risk Governance, and Public Trust
- Digital Platforms, Environmental Legitimacy, and Contestations of Power
- Environmental Media Propaganda, Misinformation, and the Politics of Ecological Representation

Overall, this special issue resonates with *Communication and the Public*'s long-standing commitment to examining how communication infrastructures shape public life, collective identities, and democratic possibilities, offering a timely and critical venue for rethinking environmental publics amid accelerating sociotechnical and ecological transformation.

Proposed Key Dates:

- Abstract Submission: March 20, 2026
- Notifications about invitations to submit full papers: March 30, 2026

*Invitations to submit a full paper do not guarantee final publication, as full manuscripts will undergo the journal's usual peer review process.

- Submission of full paper: July 31, 2026
- This special issue is planned for online publication in 2027

Please submit your abstract (maximum 500 words) in English to all guest editors, Dr. Dechun Zhang (dezh@hum.ku.dk), Dr. Weiai Xu (weiaixu@umass.edu) & Dr. Han Lin (linhan741@gmail.com) with the subject line: " **CAP Special Issue Submission.**"

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